CODERED '26 | INDUSTRY CHALLENGE BRIEF

Problem Statement 1
Reverse Engineer Anything [REA]

GenAI-assisted deconstruction and reinvention of real-world applications

OBJECTIVE Leverage generative AI to deconstruct and analyse the architecture, features and user interface of an existing application, then build an upgraded, more creative version that demonstrably improves on the original.

1. Challenge Overview

Company / Partner Name: MyStartupWave

Challenge Title: Reverse Engineer Anything [REA]

Industry / Track: Generative AI / Software Engineering

Challenge Contact: MyStartupWave Challenge Desk (mystartupwave@gmail.com)

Challenge Type: Existing Product Enhancement

Brief Summary: Teams will select a widely used, publicly available application and use generative AI to systematically deconstruct its architecture, features, data flows and user interface. Using that analysis, they will build a working, upgraded version that delivers measurable improvements in functionality, usability, performance or design, and document every step of the reverse-engineering process.

2. Industry Context

Modern product teams spend a large share of their early discovery effort on competitive teardown: studying how successful applications structure navigation, onboarding, data models and monetisation. This work is manual, slow and rarely documented in a reusable form.

At the same time, generative AI models can now read screenshots, interpret public documentation, infer API contracts from observed behaviour and produce working code scaffolding. This creates an opportunity to compress weeks of product discovery and prototyping into days, provided the result is genuinely better than the original and is built ethically, without copying proprietary code or assets.

3. Problem Statement

- **Problem and who is affected:** Startups, product managers and developers need to understand, benchmark and improve on established applications, but today they rely on manual teardown, scattered screenshots and guesswork about how a product works internally.
- **When and where it occurs:** During market research, competitor analysis, MVP planning and redesign cycles, typically before any code is written, when wrong assumptions are most costly.
- **Limitations of current solutions:** Teardown blogs and UX case studies are static, shallow and quickly outdated. AI coding assistants can generate UI copies, but they produce cosmetic clones without understanding architecture, edge cases or the reasons behind design decisions. Nothing connects structured analysis to a working, improved build.
- **Why it is worth solving:** Competitive analysis is standard practice in every product team, and app clones are among the most common developer projects, yet very few clones add meaningful innovation. A

disciplined, AI-accelerated teardown-to-upgrade method has clear value for startups and for engineering education.

4. Target Users and Stakeholders

Primary users: Developers and early-stage founders building or improving a product in an established category.

Secondary users / stakeholders: Product managers, UX designers, engineering educators and innovation teams.

Current user journey or workflow: Pick a reference app → explore it manually and take screenshots → guess the data model and APIs → design a copy in Figma → build from scratch → discover gaps late. Improvements, if any, are ad hoc and never measured against the original.

Key user pain points: Teardown is slow and undocumented; architecture is invisible from the outside; clones copy weaknesses along with strengths; and there is no objective way to prove the new version is better.

5. Product Opportunity

- **What users should accomplish:** Produce a structured, evidence-backed teardown of a real application (feature map, user flows, inferred architecture, data model and API surface) with substantial help from generative AI.
- **Meaningful improvement:** Turn that analysis into a working, upgraded application that fixes the weaknesses identified and adds at least two original, non-trivial features.
- **What makes it different:** The upgrade is proven objectively through side-by-side benchmarks (task completion time, steps per task, performance, accessibility) rather than opinion, and every design decision traces back to evidence from the teardown.

6. Expected Product Capabilities

Must-have capabilities:

- AI-driven teardown report covering a feature inventory, screen-by-screen UI breakdown, user flows, inferred system architecture, data model (ER diagram) and API/endpoint map.
- Traceability: every inference in the report is labelled as observed, documented or AI-inferred, together with the evidence (screenshot, public document or network trace) that supports it.
- A functional, deployed application that reproduces the target app's core user journeys end to end (not static mock-ups).
- At least two significant original improvements (e.g., a new AI-powered capability, a redesigned workflow, offline support, an accessibility overhaul), each justified by a weakness found in the teardown.
- A side-by-side, metric-based comparison of the original and the upgraded version.
- An original codebase and original design assets, with no copied proprietary code, logos, images or copyrighted content.

Nice-to-have capabilities:

- A reusable "REA pipeline" that can analyse a second application with minimal changes (e.g., screenshot-to-component-tree, automatic flow-graph extraction).
- Automatic generation of architecture diagrams and a product requirements document (PRD) from the teardown.
- Usability test results from at least five real users.
- Web and mobile parity.

Expected user interactions: Users explore the teardown as an interactive report (feature map, flows, architecture diagrams), then use the upgraded application to complete the same core tasks as in the original, switching between the "original flow" and the "upgraded flow" to see the difference.

Expected demonstration / deliverable:

- Live walkthrough of the teardown report and the AI workflow used to produce it.
- Live demo of the deployed upgraded application completing at least three core journeys.
- Benchmark comparison of original vs. upgraded version.
- Public repository with README, architecture diagram and an AI/prompt usage log.

7. Technical and Operational Constraints

Technical or platform constraints: The target application must be publicly accessible and widely used (e.g., a leading app in its category). Analysis may rely only on legitimate sources: the public UI, public documentation, public APIs and the team's own observation of their own accounts. Decompiling, bypassing DRM or circumventing security controls is not permitted. Any technology stack may be used.

Performance, accuracy, or latency requirements: The upgraded application must match or beat the original on at least two measurable dimensions (e.g., time-to-interactive, number of steps for a core task, Lighthouse performance or accessibility score). Core pages should load in under 2.5 seconds on a standard connection. Teardown inferences must be at least 80% accurate when spot-checked by the judges.

Security, privacy, or compliance requirements: No real user data may be scraped or stored. Authentication, if implemented, must use hashed passwords or a trusted identity provider. Follow OWASP Top 10 basics and respect the target application's terms of service.

Hardware, connectivity, or deployment limitations: Must be deployed to a publicly reachable URL (or an installable build for mobile) for judging, and must run on a standard laptop and a mid-range smartphone.

Budget or resource constraints: Solutions should run on free tiers or open-source tooling. Any paid API usage is the team's responsibility and must be documented.

Domain-specific safety requirements: No feature may mislead users into believing the upgraded app is the original: no impersonation, look-alike login pages or use of the original's brand assets.

8. Available Resources and Support

Resources marked "Confirmed" are guaranteed by the partner. "Permitted" items may be used by teams at their own discretion and cost. "Under consideration" items are not yet confirmed.

APIs, SDKs, developer tools, or technology credits: Confirmed: none provided by the partner. Permitted: any generative AI APIs or open-source models the team has access to (e.g., free tiers of LLM and vision APIs).

Sample datasets or permitted data sources: The public UI of the chosen application, its official documentation and public API references. No private or personal data.

Sandbox environments or test infrastructure: Not provided. Teams use their own free-tier hosting (e.g., Vercel, Render, Firebase).

Documentation and reference materials: Public product documentation and app-store listings of the chosen application; OWASP Top 10; WCAG 2.2 accessibility guidelines.

Mentors, domain experts, or technical sessions: Confirmed: MyStartupWave engineers during the scheduled mentoring rounds (subject to the event schedule).

Other resources or support: Under consideration: a short session on ethical reverse engineering and IP boundaries.

9. Business and Adoption Context

Intended customer or beneficiary: Startups and product teams entering established markets; engineering education programmes.

Current alternatives or competing solutions: Manual competitor teardowns, UX case-study blogs, design-inspiration libraries, and AI UI generators that produce screen-level copies without architectural understanding.

Potential value created: Faster product discovery (weeks reduced to days), lower prototyping cost, better-informed product decisions and a reusable analysis method.

Potential adoption pathway: Start as an internal discovery tool for startups and studios; evolve into a SaaS "product teardown assistant" or an educational offering for engineering programmes.

Commercial or operational considerations: Generative AI inference cost per teardown, legal review of IP boundaries, and integration with design and project tools (e.g., Figma, Jira).

Known assumptions or uncertainties: Generative AI can infer architecture with acceptable accuracy from external evidence alone; the improvements chosen by teams are valued by real users.

10. Success Indicators

Functional outcomes: Complete teardown report covering all required sections; upgraded application reproduces at least three core journeys end to end with no critical bugs; at least two original improvements working live.

User or customer outcomes: In a short usability test, users complete core tasks faster or in fewer steps than in the original, and at least 70% prefer the upgraded version.

Technical performance indicators: Measurable gains on at least two metrics (e.g., Lighthouse performance/accessibility ≥ 90, fewer steps per task, faster load time); teardown inference accuracy ≥ 80% on judge spot-checks.

Business or operational indicators: A clear statement of who would use the upgraded product and why they would switch from the original.

Safety, reliability, or compliance indicators: Zero copied proprietary code or brand assets; documented compliance with the target application's terms of service.

11. Scope and Boundaries

In scope: Selection and justification of one target application; AI-assisted teardown; a working upgraded application; benchmark comparison; documentation.

Out of scope: Decompiling binaries, bypassing authentication or paywalls, exact replication of proprietary algorithms, migrating real users or data, and full feature parity with the original.

Permitted assumptions or simplifications: Teams may focus on the 3–5 most important user journeys instead of the entire application. Backend services may be simplified or mocked where they are not central to the improvement, provided this is disclosed.

Dependencies (systems, approvals, resources): Access to generative AI tools, public hosting and a publicly accessible target application.

12. Industry Engagement

Will the company provide mentors or subject-matter experts? Yes. Engineers from MyStartupWave will be available during the scheduled mentoring rounds for architecture, product and feasibility reviews.

Will teams receive feedback during development? Yes. Teams will receive structured feedback at the designated mid-event checkpoint(s), focused on scope, technical approach and progress against the success indicators.

Will the company participate in evaluation or demonstrations? Yes. MyStartupWave representatives will attend the final demonstrations and take part in evaluation alongside the CODERED '26 jury.

Is further validation, pilot discussion, or product exploration possible after the event? Possible. Teams with exceptional solutions may be invited to a follow-up discussion on further validation, mentoring or pilot exploration.

Any follow-on engagement is subject to mutual interest and separate agreement.

13. Intellectual Property and Participation Terms

The following terms apply to all participating teams and are communicated before development begins. Participation does not imply any automatic transfer of ownership or commercial rights; any subsequent arrangement must be agreed separately.

Data and resource usage conditions: Use only public information about the target application and data the team owns or has permission to use. Do not scrape personal data or violate any platform's terms of service.

Confidentiality requirements, if any: None. This brief and any materials shared for this challenge are non-confidential.

Intellectual property and ownership terms: Teams retain full ownership of the code, designs, models and other intellectual property they create. Participation does not transfer any ownership or commercial rights to MyStartupWave.

Restrictions on using company names, data, or materials: Do not use MyStartupWave's name or branding without written permission. The target application's name may be used only descriptively in the teardown; its logos, trademarks and brand assets must not appear in the upgraded product.

Terms governing subsequent pilots, licensing, investment, employment, or commercial collaboration: Any pilot, licensing, investment, employment or commercial collaboration after the event is subject to mutual interest and a separate written agreement.

14. Partner Contact and Clarifications

Primary contact: MyStartupWave Challenge Desk

Email: mystartupwave@gmail.com

Reference documentation / resource links: This challenge brief; official documentation of the chosen target application; OWASP Top 10; WCAG 2.2.

Clarification process: Send questions in writing to the challenge contact or raise them through the official CODERED '26 helpdesk. Answers that affect scope or evaluation will be shared with all teams on this challenge at the same time, to keep the competition fair.

Submission Checklist

- ☑ The problem is real, relevant, and clearly stated.
- ☑ Target users and desired outcomes are identified.
- ☑ Teams have freedom to develop their own solutions.
- ☑ A working prototype is reasonably achievable within the challenge timeline.
- ☑ Business value and adoption can be evaluated.
- ☑ Technical constraints and available resources are stated.
- ☑ Data, safety, confidentiality, and IP terms are clear.
- ☑ Evaluation expectations are communicated consistently.

THE GUIDING PRINCIPLE

The company defines the opportunity. The team defines the solution.